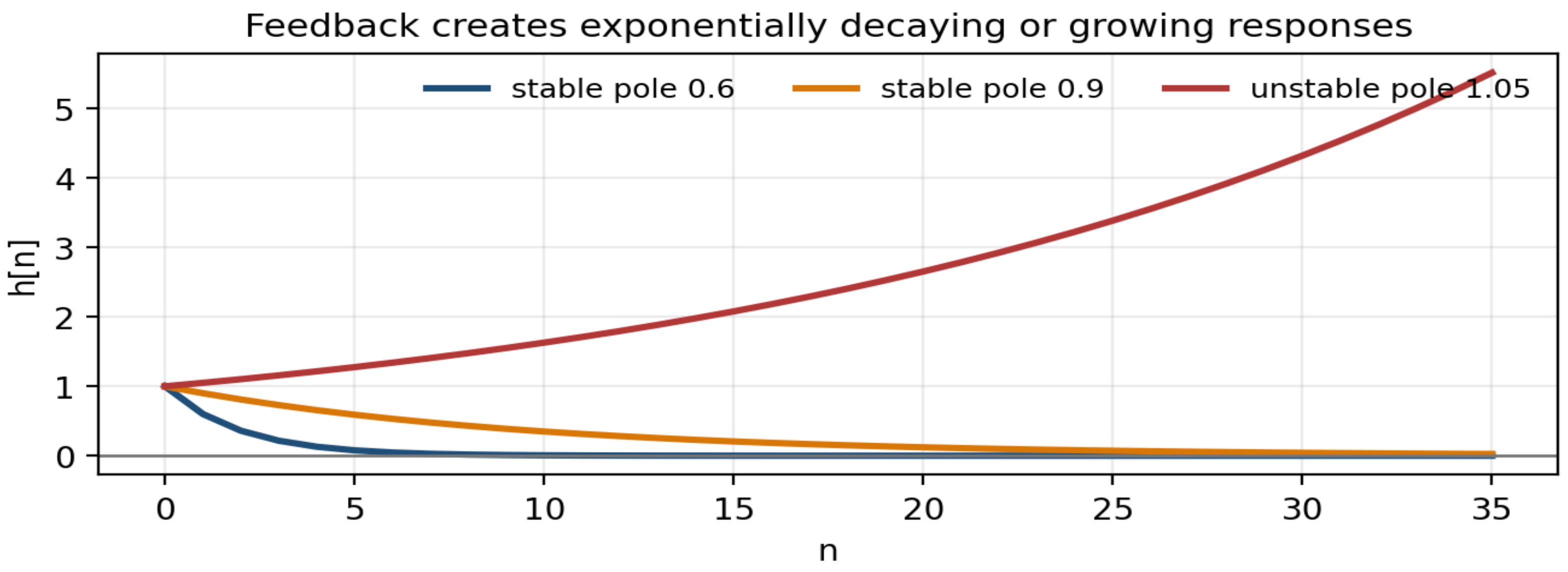
Topic 8

IIR Filters

recursive filters • poles and zeros • stability • frequency response



Key idea. IIR filters use feedback, so pole locations govern transient behaviour and stability.

IIR filters versus FIR filters

Feedback is the defining difference

- FIR: output is a finite weighted sum of present and past inputs.
- IIR: output also depends on previously computed outputs.
- Feedback can produce an infinite-duration response from a one-sample input.
- IIR filters can be very efficient, but stability must be checked.

$$\text{FIR: } y_n = \sum_{k=0}^{M} b_k x_{n-k}$$

$$\text{IIR: } y_n = \sum_{k} b_k x_{n-k} + \sum_{\ell} a_\ell y_{n-\ell}$$

finite input

infinite response tail

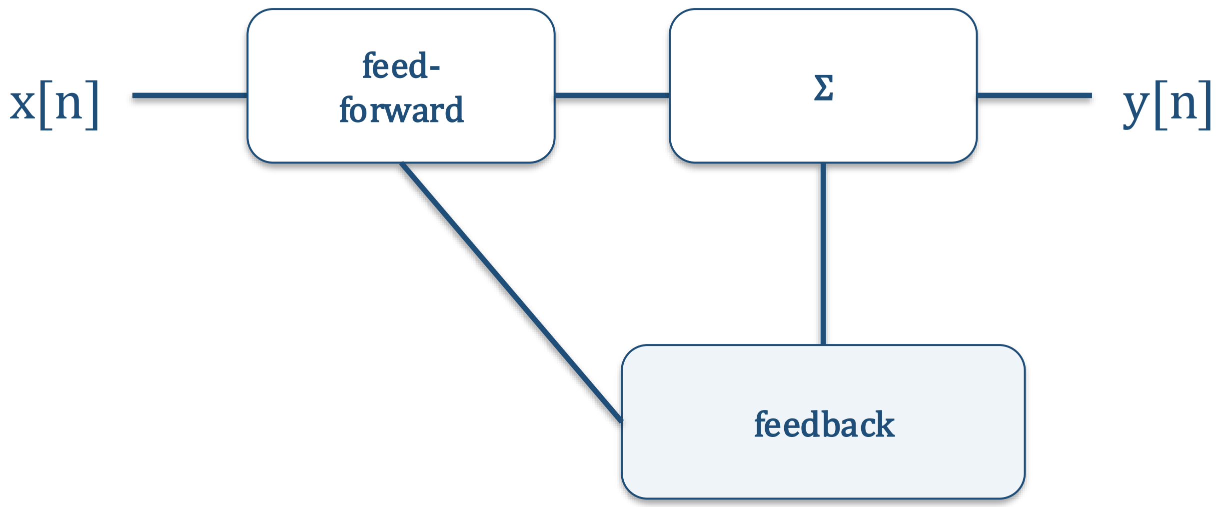
Key idea. The feedback path is what makes an IIR response “infinite.”

The general IIR difference equation

A recursive input–output rule

$$y_n = \sum_{k=0}^{M} b_k x_{n-k} + \sum_{r=1}^{N} a_r y_{n-r}$$

- b[k] weights present and past input samples.
- a[r] weights past output samples and creates feedback.
- The filter order is determined by the highest feedback/input delay used.
- If all a[r]=0, the equation reduces to an FIR filter.



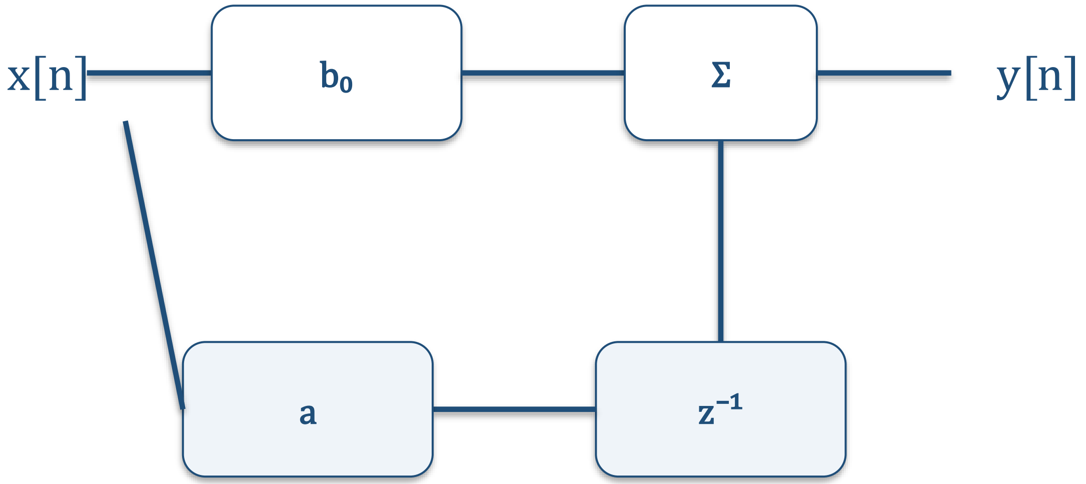
Key idea. IIR filters are algorithms with memory of their own outputs.

First-order IIR and initial rest

How recursion starts

$$y[n] = a\,y[n-1] + b_0x[n] + b_1x[n-1]$$

- At n=0 the recursion may need y[−1].
- Initial-rest conditions set x[n]=0 and y[n]=0 for n<0.
- With initial rest, the iterated IIR system is LTI.
- The output after the input ends is governed by the pole/feedback term.



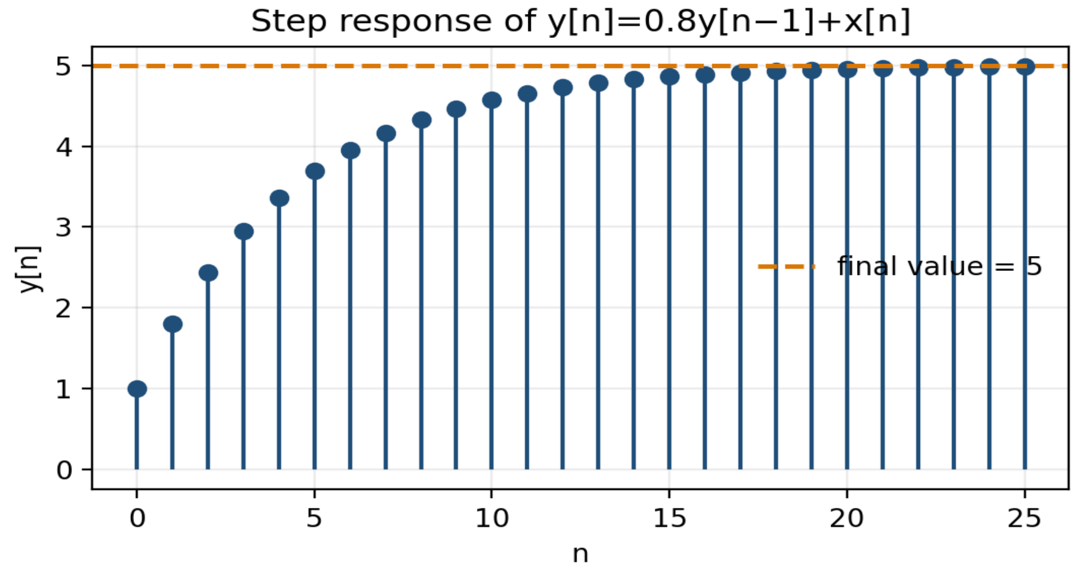
Key idea. Initial-rest conditions define a clean starting point for the recursion.

Example 1: iterating a recursion

Worked problem: step response

$$y[n] \;=\; 0.8y[n-1] \;+\; x[n], \qquad x[n] = u[n], \qquad y[-1] = 0$$

1. $n = 0$: $y[0] = 0.8 \cdot 0 + 1 = 1.$
2. $n = 1$: $y[1] = 0.8 \cdot 1 + 1 = 1.8.$
3. $n = 2$: $y[2] = 0.8 \cdot 1.8 + 1 = 2.44.$
4. $General$: $y[n] = 1 + 0.8 + \cdots + 0.8^{n} \;=\; (1 - 0.8^{n+1})/(1 - 0.8).$
5. $As\ n \to \infty, y[n] \to 1/(1 - 0.8) = 5.$



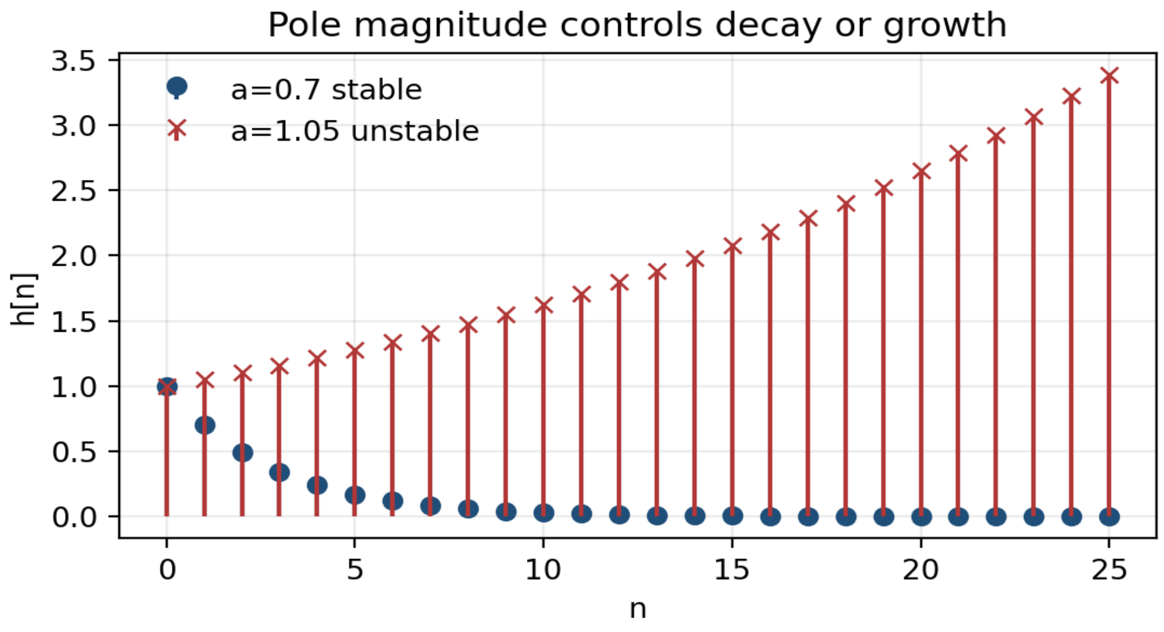
Key idea. A stable feedback coefficient accumulates a finite geometric series.

Impulse response of a first-order IIR

A one-sample input creates an infinite tail

$$H(z) = 1/(1 - az^{-1}) \quad \Leftrightarrow \quad h[n] = a^n u[n]$$

- The impulse input starts the recursion.
- After the impulse disappears, feedback keeps producing new samples.
- |a|<1 gives a decaying impulse response.
- |a|>1 gives a growing impulse response and instability.



Key idea. The pole radius controls the rate of exponential decay or growth.

System function of a first-order IIR

Recursion becomes a rational function

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

$$Y(z) = a_1 z^{-1} Y(z) + (b_0 + b_1 z^{-1}) X(z)$$

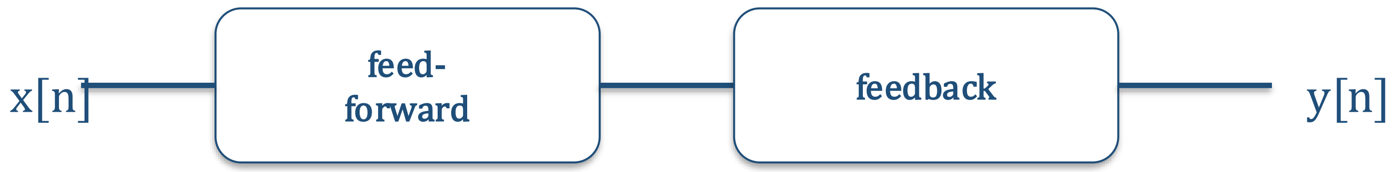
$$H(z) = Y(z)/X(z) = (b_0 + b_1 z^{-1})/(1 - a_1 z^{-1})$$

- Numerator = feed-forward coefficients.
- Denominator = feedback polynomial.
- FIR filters have polynomial H(z); IIR filters have rational H(z).

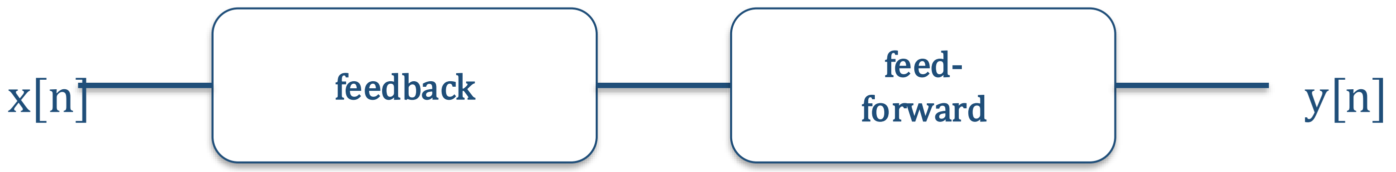
Key idea. The denominator is where poles, recursion, and stability enter.

Block-diagram structures

Direct forms implement the same H(z)



Direct form I: separate input and output delay paths



Direct form II: combines delay chains into one internal state w[n]

- Both forms realize the same system function.
- Direct form II can use fewer delay elements.
- Different diagrams are equivalent when the input-output equation is identical.

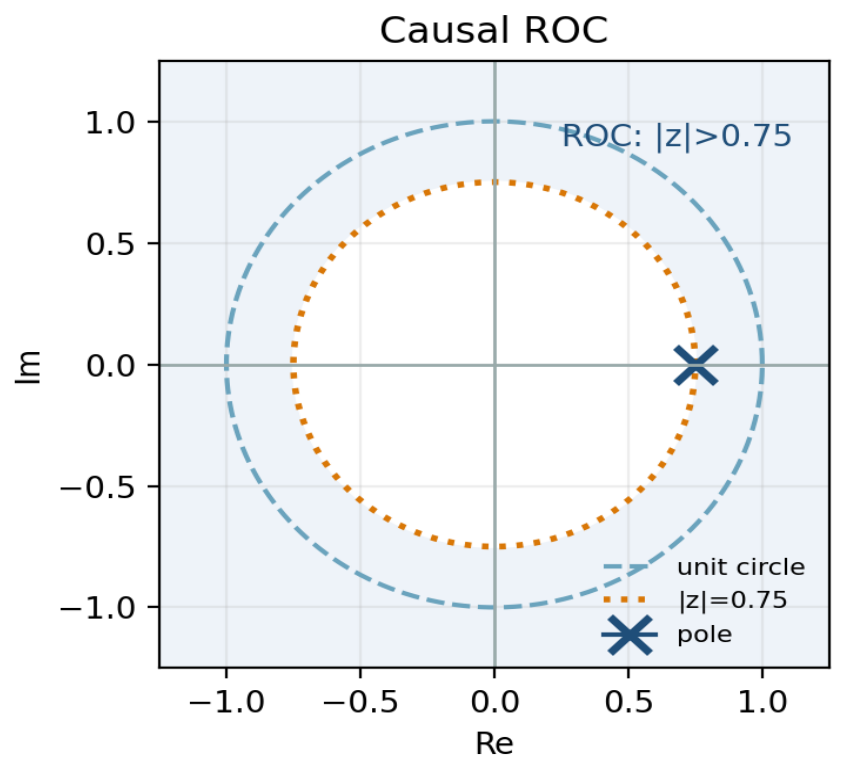
Key idea. Implementation details can change, but H(z) is the mathematical identity of the filter.

Region of convergence

When an infinite z-transform sum exists

$$\sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}}, \qquad \text{ROC: } |z| > |a|$$

- The z-transform of an IIR impulse response is an infinite geometric series.
- The region of convergence tells where that series converges.
- For a causal exponential, the ROC is outside the pole radius.
- The frequency response exists only if the unit circle lies in the ROC.



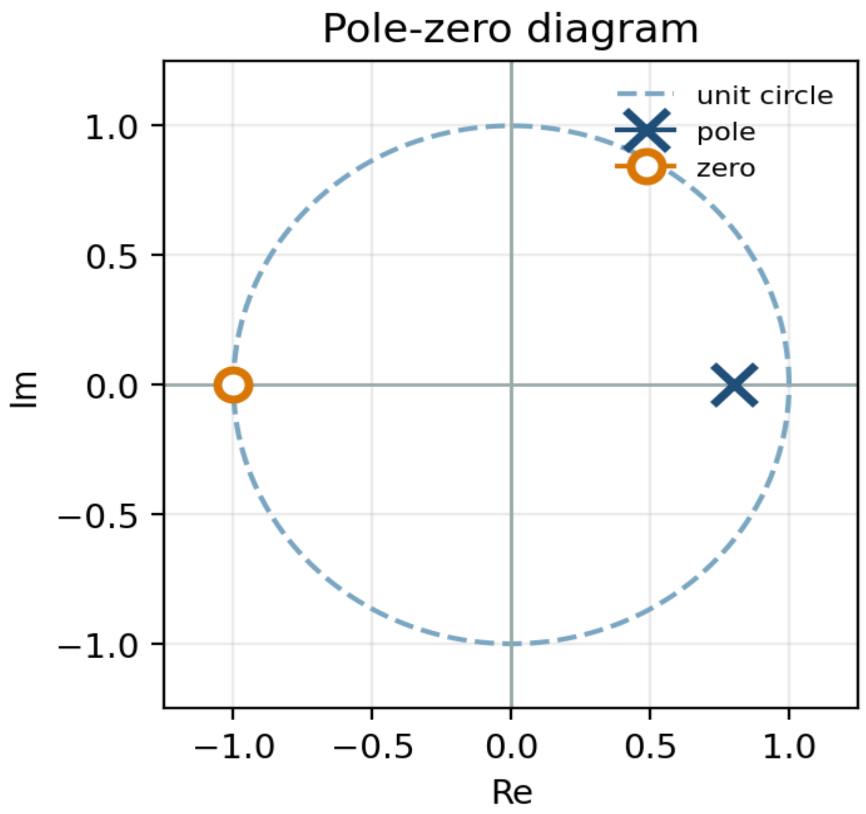
Key idea. For stable causal IIR filters, the ROC must include the unit circle.

Poles and zeros

Roots of the numerator and denominator

$$H(z) = K \cdot \Pi_i(1 - z_i z^{-1}) \,/\, \Pi_r(1 - p_r z^{-1})$$

- Zeros z_i make H(z)=0.
- Poles p_r make H(z) unbounded.
- Poles shape transients and stability.
- Zeros create attenuation or exact nulls at selected frequencies.



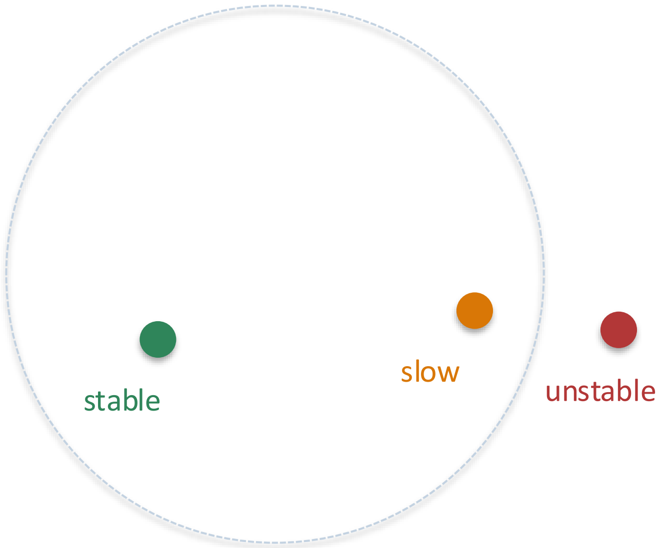
Key idea. A pole-zero diagram is a compact visual description of an IIR filter.

Pole location and stability

Causal BIBO stability criterion

$$causal\ IIR\ stable \iff all\ poles\ satisfy\ |p_r| < 1$$

- Inside the unit circle: decaying transient.
- On the unit circle: persistent oscillation or marginal case.
- Outside the unit circle: growing transient.
- Poles close to the unit circle decay slowly, so the filter rings longer.



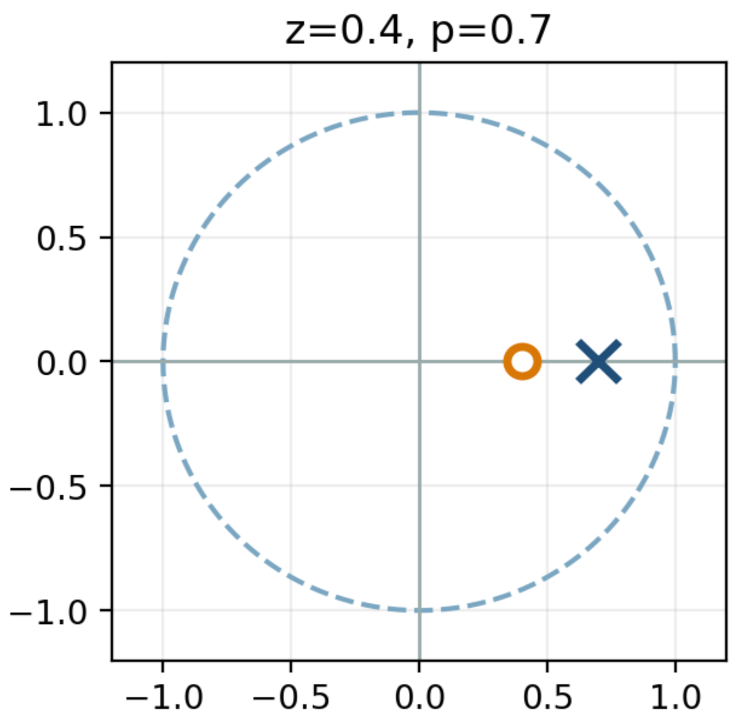
Key idea. Stability is mainly a pole-radius question.

Example 2: from equation to poles and zeros

Worked problem

$$y[n] = 0.7y[n-1] + x[n] - 0.4x[n-1]$$

1. $Take\ z - transforms\colon\ Y(z) = 0.7z^{-1}Y(z) + (1 - 0.4z^{-1})X(z).$
2. $Collect\ Y(z)\colon\ Y(z)(1 - 0.7z^{-1}) = (1 - 0.4z^{-1})X(z).$
3. $H(z) = (1 - 0.4z^{-1})/(1 - 0.7z^{-1}).$
4. $Zero\ at\ z = 0.4;\ pole\ at\ z = 0.7.$
5. $The\ causal\ system\ is\ stable\ because\ |0.7| < 1.$



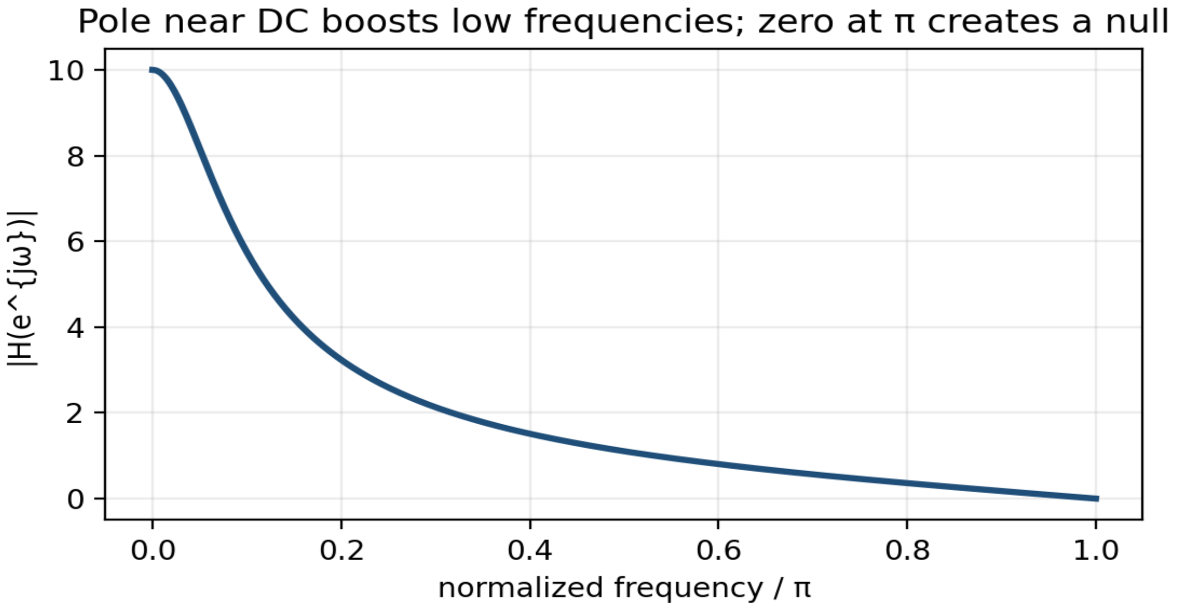
Key idea. The same algebra gives the implementation, pole-zero plot, and stability answer.

Frequency response of an IIR filter

Evaluate the system function on the unit circle

$$H(e^{j\hat{\omega}}) = H(z)|_{z = e^{j\hat{\omega}}}$$

- Requires the unit circle to lie inside the ROC.
- $|H(e^{j\hat{\omega}})|$ scales sinusoidal amplitude.
- $\angle H(e^{j\hat{\omega}})$ shifts sinusoidal phase.
- Poles near the unit circle create peaks; zeros on it create nulls.



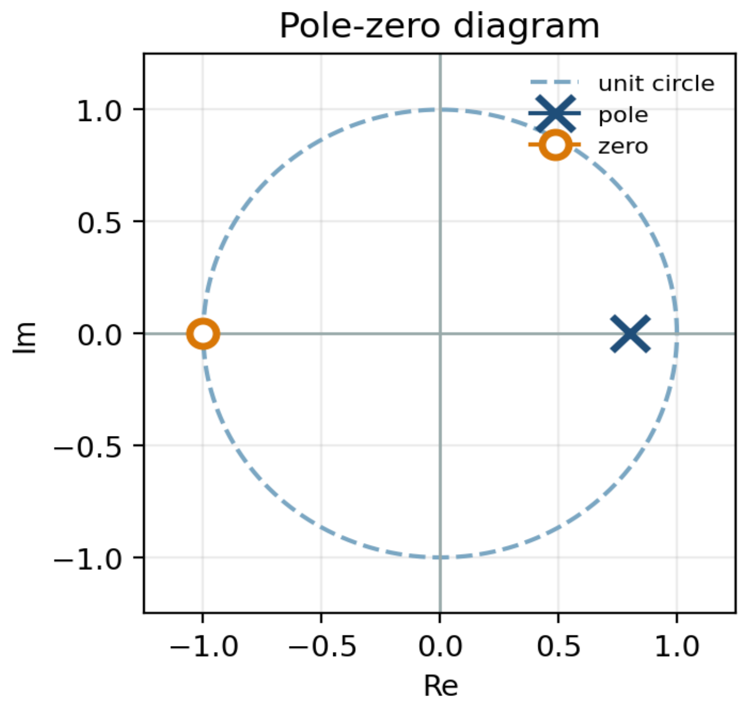
Key idea. The frequency response is the value of H(z) traced around the unit circle.

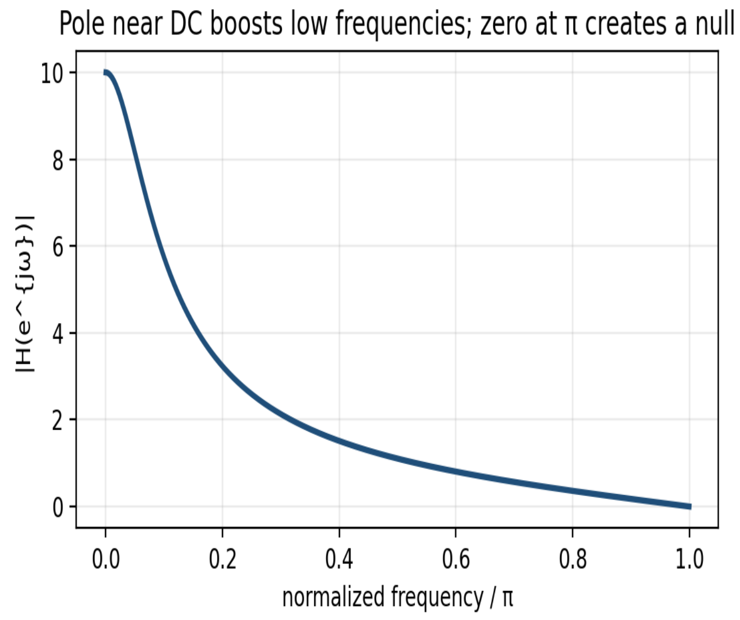
How poles and zeros shape |H(e^{jω̂})|

Distance intuition on the unit circle

$$|H(e^\wedge\{j\hat{\omega}\})| = |numerator\ distances| \,/\, |denominator\ distances|$$

- A unit-circle point close to a pole gives large gain.
- A unit-circle point close to a zero gives small gain.
- A zero exactly on the unit circle gives an exact frequency null.
- This geometric view explains notches, resonances, and low-pass behaviour.

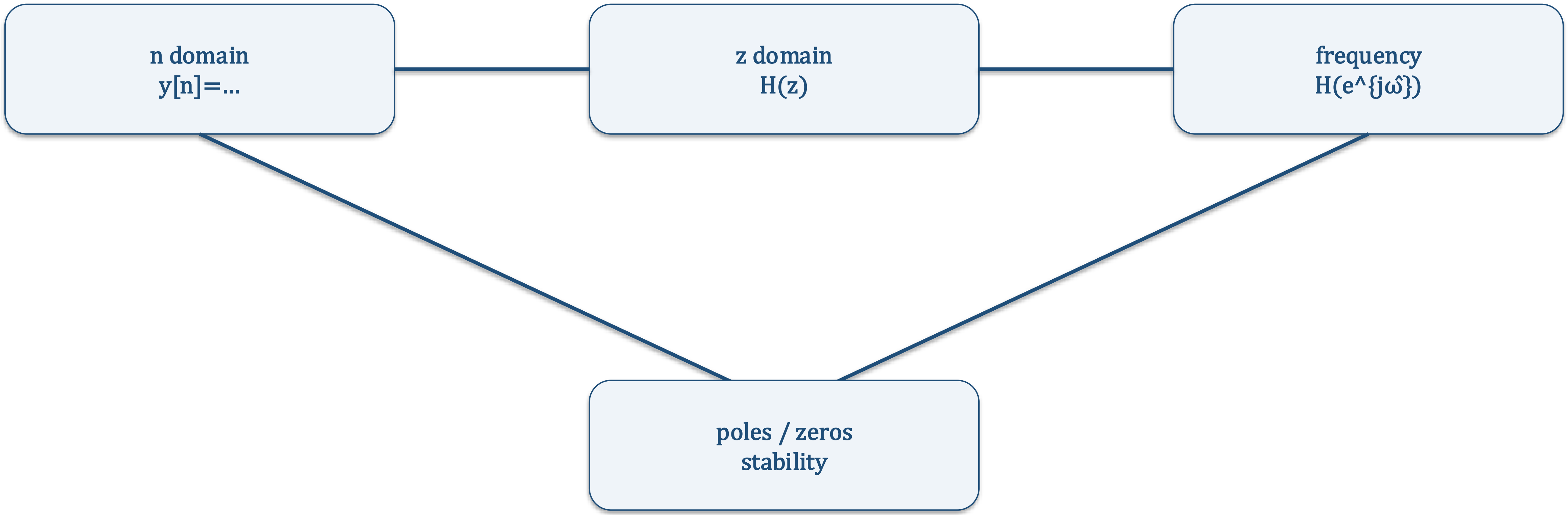




Key idea. Frequency selectivity is geometry: poles pull the response up, zeros push it down.

The three domains

Equivalent views of the same IIR filter



- Time domain: recursion, impulse response, transient.
- z domain: rational function, poles, zeros, ROC.
- Frequency domain: gain and phase for sinusoids.

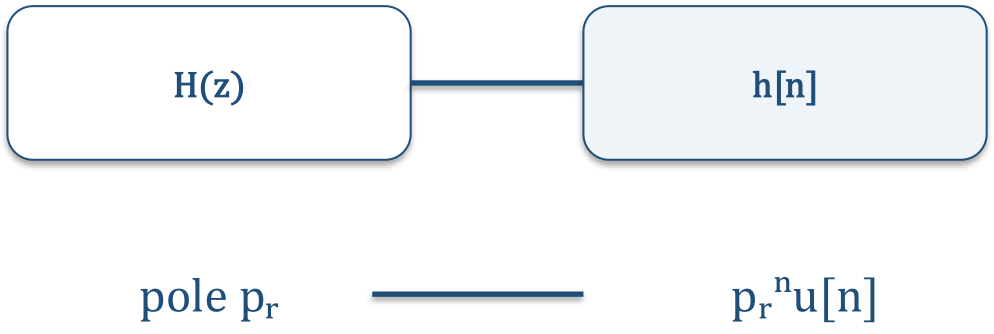
Key idea. Move between domains to choose the easiest description for the question.

Inverse z-transform

From a rational function back to a sequence

$$H(z) \;=\; \Sigma\; residues\; of\; 1/(1 - p_r z^{-1}) \;\;\Rightarrow\;\; h[n] = \Sigma\; C_r p_r^{n} u[n]$$

- Factor the denominator and identify poles.
- Use partial-fraction expansion.
- Apply the transform pair $p^n u[n] \Leftrightarrow 1/(1-pz^{-1})$.
- When numerator degree is too high, first perform polynomial division.



Key idea. Partial fractions turn rational H(z) into sums of causal exponentials.

Example 3: inverse z-transform

Worked problem

$$Y(z) = 1/[(1 - 0.5z^{-1})(1 - 0.8z^{-1})]$$

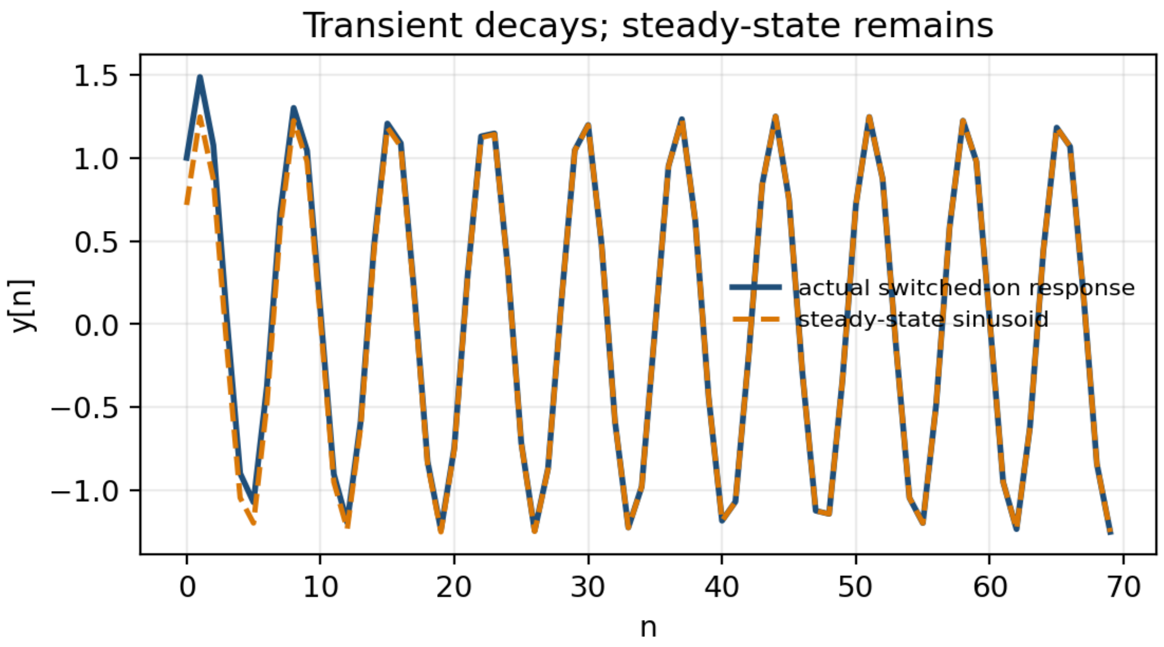
1. $Write\ Y(z) = A/(1 - 0.5z^{-1}) + B/(1 - 0.8z^{-1}).$
2. $Match\ coefficients\colon\ A + B = 1\ and\ 0.8A + 0.5B = 0.$
3. $Solve\colon\ A = -5/3\ and\ B = 8/3.$
4. $Use\ p^{n}u[n]\ \Leftrightarrow\ 1/(1 - pz^{-1}).$
5. $y[n] = -(5/3)(0.5)^{n}u[n] + (8/3)(0.8)^{n}u[n].$

Transient and steady-state response

A switched-on sinusoid is not initially an eigenfunction

$$x[n] = e^{j\hat{\omega}_0 n}u[n] \;\rightarrow\; y[n] = \mathit{transient} + H(e^{j\hat{\omega}_0})e^{j\hat{\omega}_0 n}$$

- The forced term is the sinusoidal steady state.
- The natural term depends on the pole locations.
- For a stable filter, pole-dependent transients decay.
- If a pole is outside the unit circle, the transient grows and dominates.



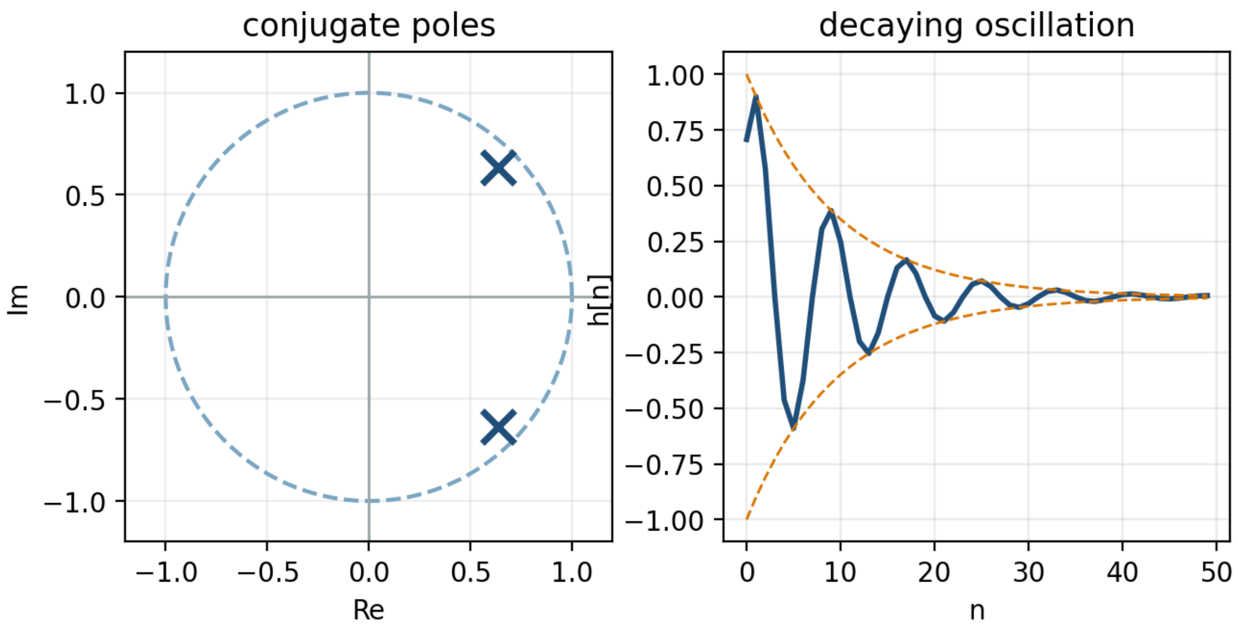
Key idea. Stable IIR filters settle to the frequency-response prediction after the transient decays.

Second-order IIR filters

Complex-conjugate poles create resonances

$$H(z) = B(z)/(1 - 2r\ cos\theta\ z^{-1} + r^2 z^{-2})$$

- Conjugate poles: $p_{1,2} = re^{\{\pm j\theta\}}$.
- Pole angle θ sets oscillation frequency.
- Pole radius r sets decay time and resonance sharpness.
- Stability requires r<1 for the causal system.



Key idea. Second-order sections are the building blocks of resonators and many practical IIR filters.

Example 4: design a damped oscillator

$$desired\ poles:\ \ p_{1,2} = 0.9e\hat{}\{\pm j\pi/4\}$$

1. $Use\ denominator\ 1 - (p_1 + p_2)z^{-1} + p_1 p_2 z^{-2}.$
2. $p_1 + p_2 = 2r\ cos\theta = 2 \cdot 0.9 \cdot cos(\pi/4) \approx 1.273.$
3. $p_1 p_2 = r^2 = 0.81.$
4. $A(z) = 1 - 1.273z^{-1} + 0.81z^{-2}.$
5. The impulse response oscillates at $\pi/4$ rad/sample and decays like 0.9^n.

Implementation convention

Numerator bb and denominator aa

scipy.signal.lfilter(bb, aa, xx)

- bb contains feed-forward coefficients: b[0], b[1], …
- aa contains denominator coefficients: 1, −a[1], −a[2], … under this sign convention.
- The same convention appears in many filter-design functions.
- Always check signs when translating from a difference equation to code.

$$bb = [b_0, b_1]$$

$$aa = [1, -a_1]$$

filter

Key idea. Implementation errors often come from denominator-sign conventions.

Common interpretation errors

Checks before trusting an IIR analysis

- Forgetting initial-rest conditions when iterating the recursion.
- Treating an IIR impulse response as finite because only a few samples are plotted.
- Checking zeros but ignoring poles when deciding stability.
- Evaluating $H(e^{j\hat{\omega}})$ when the unit circle is not in the ROC.
- Confusing transient response with steady-state frequency response.

Key idea. For IIR filters, poles are the first thing to check.

Notation

Symbol	Meaning
x[n], y[n]	Input and output discrete-time sequences
a_k	Recursive (feedback) filter coefficients
b_k	Non-recursive (feed-forward) filter coefficients
h[n]	Impulse response
H(z)	System function or transfer function
X(z), Y(z)	z-transforms of the input and output
z	Complex z-plane variable
ROC	Region of convergence of a z-transform
p_k, z_k	Pole and zero locations
H(e^{jω̂})	Frequency response evaluated on the unit circle
ω̂	Normalized radian frequency in radians per sample
u[n], δ[n]	Unit-step and unit-impulse sequences
r, θ	Radius and angle of a complex-conjugate pole pair